PROJET JAVA

Statistiques NBA

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01 PRÉSENTATION

Aperçu du site et des différents fichiers

PRÉSENTATION

https://www.nbastuffer.com/2023-2024-nba-player-stats/

2023-2024 NBA Player Stats | Regular Season

RANK	NAME	TEAM	POS	AGE	GP	MPG	USG%	TO%	FTA	FT%	2PA	2P%	3PA	3P%	eFG%	TS%	PPG	RPG	APG	SPG	BPG	TPG	P+R	P+A	P+R+A	VI	ORtg	DRtg
1	Joel Embiid	Phi	C	29.8	22	33.9	37.4	15.2	254	0.89	399	0.569	71	0.338	0.56	0.646	34.2	11.7	6	1.1	1.9	3.8	45.9	40.2	51.9	15.8	127.7	102.9
2	Luka Doncic	Dal	G	24.8	24	36.7	35.3	14.9	190	0.8	310	0.574	243	0.379	0.571	0.616	32.7	8.3	9.1	1.3	0.5	4	41	41.8	50.1	14.7	121.4	111.6
3	Giannis Antetokounmpo	Mil	F	29	24	34.6	33.8	16.6	273	0.674	413	0.663	40	0.225	0.635	0.662	31.6	10.8	5	1.5	1.2	3.8	42.4	36.7	47.4	13.8	125.3	107.1
4	Kevin Durant	Pho	F	35.2	21	36.7	32.3	14.2	175	0.88	325	0.529	102	0.48	0.575	0.64	30.7	6.3	5.6	0.7	1.2	3.4	37	36.3	42.6	11.2	122.3	113.4
5	Shai Gilgeous-Alexander	Okc	G	25.4	23	35	32.2	9	183	0.913	402	0.582	74	0.324	0.567	0.635	30.7	5.5	6.5	2.7	0.8	2	36.3	37.2	42.7	11.8	130	107.7
6	De'Aaron Fox	Sac	G	26	18	35.5	33.7	10.3	142	0.732	250	0.528	150	0.387	0.548	0.586	30.1	4.4	6.4	1.4	0.3	2.6	34.5	36.5	40.9	10.7	118.3	110.6
7	Stephen Curry	Gol	G	35.8	23	33.8	31	14.7	148	0.939	167	0.545	272	0.423	0.6	0.661	29	4.9	4.2	0.7	0.3	3.3	33.9	33.2	38.1	10	123.2	114
8	Devin Booker	Pho	G	27.1	16	35.2	31.8	13.8	133	0.917	228	0.509	78	0.41	0.536	0.617	28.1	5.9	8.3	0.7	0.4	3.1	34	36.4	42.3	12.6	126.2	109.5
9	Trae Young	Atl	G	25.2	23	35.9	32.1	16.9	188	0.872	264	0.466	206	0.379	0.511	0.583	28	2.9	10.8	1.3	0.2	4.2	30.9	38.8	41.7	10.7	120.7	120.6
10	Jayson Tatum	Bos	F	25.8	24	36.9	30.2	13.9	164	0.793	262	0.603	208	0.341	0.563	0.608	27.5	8.8	4.2	1	0.5	3.2	36.2	31.6	40.4	10.9	116.5	105.4
11	Donovan Mitchell	Cle	G	27.3	21	36.1	31.6	11.3	127	0.89	258	0.519	183	0.35	0.522	0.577	27.3	5.8	5.5	1.9	0.4	2.7	33.1	32.8	38.6	10.6	116.5	109.5
12	Nikola Jokic	Den	C	28.8	26	33.5	31.8	12.2	164	0.787	406	0.596	97	0.309	0.571	0.611	27	12.5	9.5	1.2	0.8	2.7	39.5	36.6	49	17.6	129	105
13	Anfernee Simons	Por	G	24.5	6	32.7	31	11.5	32	0.969	61	0.508	54	0.389	0.543	0.604	26	3	5.3	1	0.2	2.5	29	31.3	34.3	9.1	122.3	115
14	Tyrese Haliburton	Ind	G	23.8	21	35.5	25.3	12.8	97	0.876	190	0.589	177	0.435	0.62	0.659	25.7	4.1	12	0.9	0.7	2.6	29.9	37.7	41.8	12.2	136.1	123.7
15	Tyrese Maxey	Phi	G	23.1	24	36.9	25.8	6.5	129	0.891	265	0.513	192	0.396	0.547	0.599	25.6	3.9	6.7	0.9	0.6	1.3	29.5	32.3	36.2	9.5	130.1	111.6

ITEMS EXTRAITS

01 Nom du joueur

02 Âge

03 Point par match

04 Rebond par match

05 Assist par match

06 Steal par match

07 Block par match

08 Poste

09 Équipe

10 Nombre de match

11 Temps de jeu

12 Nombre de trois points

+ ITEMS D'ANALYSE +

01 Polyvalence

02 Moyenne de temps de jeu

03 Âge jouant le plus

04 Corrélation - Points et Trois points

05 Régression - Steal et Block

02

DÉMONSTRATION

Petite démonstration du code

03 SCAPING

Récupération des données

SCRAPING

```
<tr class="row-2 even">
<td class="column-1"></td>
<td class="column-2">LeBron James</td>
<td class="column-3">Lal</td>
<td class="column-4">F</td>
<td class="column-5">39</td>
<td class="column-6">23</td>
<td class="column-7">35.2</td>
<td class="column-8">28.3</td>
<td class="column-9">16.6</td>
<td class="column-10">130</td>
<td class="column-11">0.715</td>
<td class="column-12">265</td>
<td class="column-13">0.623</td>
<td class="column-14">131</td>
<td class="column-15">0.405</td>
<td class="column-16">0.617</td>
<td class="column-17">0.642</td>
<td class="column-18">25.3</td>
<td class="column-19">7.5</td>
<td class="column-20">6.7</td>
...
</tr>
...
```

```
<tr class="row-3 odd">
<td class="column-1"></td>
<td class="column-2">Chris Paul</td>
<td class="column-3">Gol</td>
<td class="column-4">G</td>
<td class="column-5">38.6</td>
<td class="column-6">21</td>
<td class="column-7">26.8</td>
<td class="column-8">15</td>
<td class="column-9">13.2</td>
<td class="column-10">33</td>
<td class="column-11">0.909</td>
<td class="column-12">89</td>
<td class="column-13">0.461</td>
<td class="column-14">75</td>
<td class="column-15">0.347</td>
<td class="column-16">0.488</td>
<td class="column-17">0.532</td>
<td class="column-18">9</td>
<td class="column-19">3.6</td>
<td class="column-20">7.2</td>
...
</tr>
...
```

SCRAPING

```
public final class ItemsScrapper {

 ArrayList<Item> parseSource(String pageSource) {
   ArrayList<Item> itemList = new ArrayList<>();
   int startIndex = 0;
   int endIndex;

   while ((startIndex = pageSource.indexOf("<tr", startIndex)) != -1) {
     endIndex = pageSource.indexOf("</tr>", startIndex);
     if (endIndex != -1) {

       String itemContent = pageSource.substring(startIndex, endIndex);

       String nomJoueur = extraireInfo(itemContent, "column-2");
       Double age = parseDouble(extraireInfo(itemContent, "column-5"));
       Double pointsParGame = parseDouble(extraireInfo(itemContent,
"column-18"));
...

       String equipe = extraireInfo(itemContent, "column-3");
       Integer nbJeuxJoues = parseInt(extraireInfo(itemContent, "column-6"));
       Integer troisPoints = parseInt(extraireInfo(itemContent, "column-14"));
```

SCRAPING

```java
private String extraireInfo(String itemContent, String columnName) {
 String startTag = "<td class=\"" + columnName + "\">";
 String endTag = "</td>";

 int startIndex = itemContent.indexOf(startTag);
 if (startIndex != -1) {
   startIndex += startTag.length();
   int endIndex = itemContent.indexOf(endTag, startIndex);
   if (endIndex != -1) {
     return itemContent.substring(startIndex, endIndex).trim();
   }
 }
 return null;
}
```

SCRAPING

```
private Double parseDouble(String value) {
 try {
   return value != null ? Double.parseDouble(value) : null;
 } catch (NumberFormatException e) {
   return null;
 }
}

private Integer parseInt(String value) {
 try {
   return value != null ? Integer.parseInt(value) : null;
 } catch (NumberFormatException e) {
   return null;
 }
}
```

SCRAPING

```
        Item item = new Item();
        item.setNom_joueur(nomJoueur);
        item.setAge(age);
        item.setPoint_par_game(pointsParGame);
        item.setRebond_par_game(rebondsParGame);
        item.setAssist_par_game(assistsParGame);
        item.setSteal_par_game(stealsParGame);
        item.setBlock_par_game(blocksParGame);
        item.setPoste(poste);
        item.setTeam(equipe);
        item.setNb_game_jouer(nbJeuxJoues);
        item.setNb_minute_jouer_par_game(minutesParJeu);
        item.setTrois_points(troisPoints);


        itemList.add(item);

        startIndex = endIndex + 1;
    }
}

return itemList;
```

04 ANALYSE

Analyse des données

ANALYSE

```
public final class ItemAnalyzer {
 private ArrayList<Item> items = new ArrayList<>();

 public ItemAnalyzer(ArrayList<Item> items) {
   this.items=items;
 }
```

ANALYSE

```
public Double getMoyenne_temps_jeu() {
 double sommeTempsJeu = 0.0;

 for (Item item : items) {
   if (item.getNb_minute_jouer_par_game() != null) {
     sommeTempsJeu += item.getNb_minute_jouer_par_game();
   }
 }

 return sommeTempsJeu / items.size();
}
```

ANALYSE

```java
public List<List<String>> getPolyvalence() {
 List<List<String>> polyvalentPlayers = new ArrayList<>();

 for (Item item : items) {
   if (item.getPoint_par_game() != null &&
         item.getAssist_par_game() != null &&
         item.getRebond_par_game() != null) {
     if (item.getPoint_par_game() > 5 &&
         item.getAssist_par_game() > 2 &&
         item.getRebond_par_game() > 2) {
       ArrayList<String> playerInfo = new ArrayList<>();
       playerInfo.add(item.getNom_joueur());
       polyvalentPlayers.add(playerInfo);
     }
   }
 }

 return polyvalentPlayers;
}
```

ANALYSE

Corrélation : (Pearson)

$$r = \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{\sqrt{[\, n\Sigma x^2 - (\Sigma x)^2 \,][\, n\Sigma y^2 - (\Sigma y)^2 \,]}}$$

Régression :

$$a = \frac{n(\sum XY) - (\sum X)(\sum Y)}{n(\sum X^2) - (\sum X)^2}$$

ANALYSE

```java
public Double getCorrelation_pts_par_game_trois_points() {

 double sumPtsParGame = 0.0;
 double sumTroisPoints = 0.0;
 double sumPtsParGameTroisPoints = 0.0;
 double sumPtsParGameSquared = 0.0;
 double sumTroisPointsSquared = 0.0;

 for (Item item : items) {
   if (item.getPoint_par_game() != null && item.getTrois_points() != null) {
     double ptsParGame = item.getPoint_par_game();
     double troisPoints = item.getTrois_points();

     sumPtsParGame += ptsParGame;
     sumTroisPoints += troisPoints;
     sumPtsParGameTroisPoints += ptsParGame * troisPoints;
     sumPtsParGameSquared += Math.pow(ptsParGame, 2);
     sumTroisPointsSquared += Math.pow(troisPoints, 2);
   }
 }

 int n = items.size();

 double numerator = (n * sumPtsParGameTroisPoints) - (sumPtsParGame * sumTroisPoints);
 double denominator1 = Math.sqrt((n * sumPtsParGameSquared) - Math.pow(sumPtsParGame, 2));
 double denominator2 = Math.sqrt((n * sumTroisPointsSquared) - Math.pow(sumTroisPoints, 2));

 return numerator / (denominator1 * denominator2);
}
```

ANALYSE

```java
public Double getRegression_steal_block_par_game() {
 if (items.isEmpty()) {
   return null;
 }

 double sumSteals = 0.0;
 double sumBlocks = 0.0;
 double sumStealsBlocks = 0.0;
 double sumStealsSquared = 0.0;

 for (Item item : items) {
   if (item.getSteal_par_game() != null && item.getBlock_par_game() != null) {
     double steals = item.getSteal_par_game();
     double blocks = item.getBlock_par_game();

     sumSteals += steals;
     sumBlocks += blocks;
     sumStealsBlocks += steals * blocks;
     sumStealsSquared += Math.pow(steals, 2);
   }
 }

 int n = items.size();

 double numerator = (n * sumStealsBlocks) - (sumSteals * sumBlocks);
 double denominator = (n * sumStealsSquared) - Math.pow(sumSteals, 2);

 return numerator / denominator;
}
```

CONCLUSION

01 Polyvalence : [[LeBron James], [Chris Paul], [Kyle Lowry], [Kevin Durant], ...]

02 Moyenne de temps de jeu : ~18 minutes

03 Âge jouant le plus : ~24.5

04 Corrélation - Points et Trois points : ~0.75

05 Régression - Steal et Block : ~0.35

MERCI

EXPOSÉ RÉALISÉ
PAR
JELASSI Bader
ET
DU Alexandre